

REVIEW



A meta-analysis of serum lipid profiles in premature ovarian insufficiency



BIOGRAPHY

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KEY MESSAGE

This meta-analysis reveals that unfavourable lipids, such as serum total cholesterol, low-density lipoprotein cholesterol and triglyceride levels are significantly higher in patients with premature ovarian insufficiency, which may further contribute to the high risk of cardiovascular diseases.

ABSTRACT

Premature ovarian insufficiency (POI), defined as loss of normal ovarian functions before the age of 40 years, occurs in at least 1% of all women. It affects the reproductive system and causes many health problems and psychological stress. Abnormal serum lipid profile leads to cardiovascular diseases, which are strongly associated with high mortality in patients with POI. To date, several studies have examined the levels of different serum lipids in patients with POI. The results, however, are either inconclusive or inconsistent. Therefore, the aim of this meta-analysis was to measure whether serum levels of total cholesterol, high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C) and triglyceride are varied in patients with POI. Ten studies in total were included in this meta-analysis involving 1009 individuals: 458 patients with POI and 551 controls. Our analysis results showed that serum total cholesterol ($P < 0.00001$), LDL-C ($P < 0.0001$), and triglyceride ($P = 0.01$) levels were significantly higher in patients with POI compared with healthy controls. Serum HDL-C levels, however, did not vary significantly between controls and patients with POI. These results suggest that elevations in unfavourable lipids may contribute to the high risk of cardiovascular diseases that are observed in patients with POI.

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KEYWORDS

Cholesterol
HDL-C
LDL-C
Meta-analysis
POI
Triglyceride

INTRODUCTION

Premature ovarian insufficiency (POI), also known as premature ovarian failure (POF), is a cause of infertility that affects about 1% of women. According to the European Society of Human Reproduction and Embryology guideline (ESHRE, 2016), POI is defined by oligomenorrhoea and amenorrhoea for at least 4 months, and elevated FSH level (>25 IU/l) on two occasions more than 4 weeks apart in women younger than 40 years of age. Premature ovarian insufficiency can have significant physiological, psychological, social and economic effects and is associated with earlier mortality (Taylor, 2001; Rocca et al., 2006). Although several factors have been reported to be involved in the pathogenesis of POI, the underlying cause is not identified in most cases (Goswami and Conway, 2005).

Cardiovascular diseases are the leading causes of death in women. An increased risk of cardiovascular diseases is observed in women undergoing prophylactic bilateral oophorectomy before menopause (Rivera et al., 2009). Women with POI have been reported to be at higher risk of cardiovascular diseases than healthy controls (Roeters van Lennep et al., 2016). These observations may be attributed to the loss of ovarian functions and subsequent deficiency of endogenous oestrogen. Abnormal lipid profiles are strongly associated with the risk of cardiovascular diseases. It is known that lower levels of high-density lipoprotein cholesterol (HDL-C) and higher levels of total cholesterol, low-density lipoprotein cholesterol (LDL-C) and triglyceride are associated with an increased risk of cardiovascular diseases (Lemieux et al., 2001). To date, several studies have examined the lipid profiles in women with POI, with contradictory results. Therefore, in the present study, a meta-analysis was conducted to clarify whether serum lipid profiles, including total cholesterol, HDL-C, LDL-C and triglyceride levels, are varied in patients with POI.

MATERIALS AND METHODS

Guideline selection and data sources

The scientific community has established different guidelines to ensure the transparency and accuracy of reporting medical research (Johansen and Thomsen, 2016). The PRISMA

(Preferred reporting items for systematic reviews and meta-analyses) guideline was developed for the reporting of meta-analysis and is widely accepted and followed (Moher et al., 2009). Therefore, the PRISMA guideline was followed in the present study. PubMed, the Web of Science and Embase databases were used to search for published studies. The following keywords and strategies were used: (lipids OR triglyceride OR TG OR total cholesterol OR TC OR high-density lipoprotein cholesterol OR HDL-C OR low-density lipoprotein cholesterol OR LDL-C) AND (POI OR premature ovarian insufficiency OR POF OR premature ovarian failure). Potentially relevant studies published before March 2021 were identified by two authors (ZW and LF). In addition, the references cited in the relevant studies were screened to ensure that no related study was missed.

Inclusion and exclusion criteria

The inclusion criteria were as follows: lipid serum levels provided for patients with POI and controls; and participants had no metabolic-related diseases and did not receive hormone therapy. The exclusion criteria were as follows: participants with chromosomal abnormalities, as it can cause variation in serum lipid levels (Pirruccello and Kathiresan, 2010); levels of serum lipids measured in patients aged 40 years or older; animal study; and abstracts, letters and conference presentations.

Data extraction and quality assessment

Two investigators (ZW and LF) conducted data extraction and quality assessment independently. The following data were extracted from all included studies: name of the first author; the year of publication; and the number, age and body mass index (BMI) of patients with POI and controls. The Newcastle–Ottawa Quality Assessment Scale (NOS) (Deeks et al., 2003; Stang, 2010) was used to estimate the methodological quality of the selected studies. Scores ranging from 0 (the worst) to 9 (the best) were assigned based on the quality of the studies. The study with NOS scores lower than 6 was considered to be low quality and excluded.

Statistical analysis

The odds ratio, relative risk or risk difference are commonly used effect sizes when the outcome of interest

is a dichotomous variable. When the outcome is a continuous variable, the effect size can be represented as either the mean difference or the standardized mean difference (SMD) (Higgins et al., 2008). In the meta-analysis, the SMD has been shown to be more generalizable than the mean difference (Takeshima et al., 2014). The SMD represents the group difference relative to the pooled variability observed among groups. Because of the heterogeneity in assays used for measurements from different laboratories, to assess the changes of serum lipid levels between controls and patients with POI, an SMD with the 95% confidence interval was calculated. Statistically, the SMD magnitudes of 0.2, 0.5 and 0.8 can be interpreted as small, medium and large effect sizes, respectively (Cohen, 1988). The heterogeneity among the included studies was estimated by the Q-test and the I^2 test statistic. $P < 0.05$ and $I^2 > 50\%$ were considered to indicate significant heterogeneity (Higgins et al., 2002). The random-effects model was applied if significant heterogeneity was detected; otherwise, the fixed-effects model was used. A sensitivity analysis, which removed one study at a time and recalculated the stability of the results, was conducted to test for the robustness of the summary effect. A Begg's funnel plot was used to analyse publication bias (Begg and Mazumdar, 1994). STATA (version 15.0) and Review Manager (version 5.3) software were used for statistical analyses.

RESULTS

Characteristics of included studies

A total of 113 studies were initially identified by searching three different databases (FIGURE 1). After the duplication check, 20 studies were omitted. After the title and abstract screening of the remaining 93 studies, 81 were further excluded for not conforming to our inclusion criteria. Full-text of the remaining 12 studies were then evaluated for their eligibility. Two studies were excluded because the levels of serum lipids were measured in patients older than 40 years (Gunning et al., 2019; 2020). Finally, 10 selected articles, including 1009 individuals with 458 patients with POI and 551 controls, were processed for the meta-analyses (Senoz et al., 1996; Kalantaridou et al., 2004; Knauff et al., 2008; Gulhan et al., 2012; Kebapcilar et al., 2013;

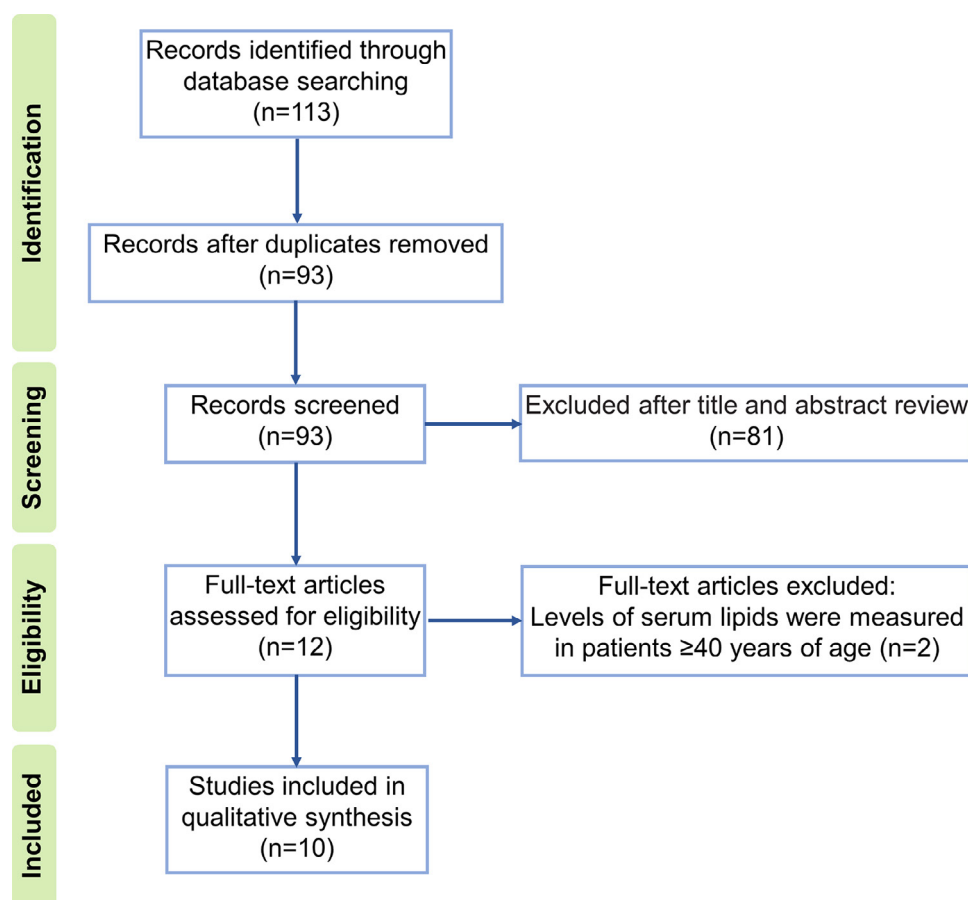


FIGURE 1 The study selection process.

Kulaksizoglu et al., 2013; Ates et al., 2014; Agacayak et al., 2016; Podfigurna et al., 2018; Bozkaya et al., 2020). The quality assessments of individual studies following the NOS are presented in **TABLE 1**. The specific lipid profiles of all the studies are presented in **TABLES 2–5**.

Serum total cholesterol, high-density lipoprotein cholesterol and low-density lipoprotein cholesterol levels in controls and patients with premature ovarian insufficiency

A meta-analysis was initially conducted using a fixed-effects model. Significant

heterogeneity among the results of serum total cholesterol levels from 10 included studies was observed ($I^2 > 50\%$, $P < 0.0001$). Therefore, a sensitivity analysis was conducted to identify the study that contributed to the significant heterogeneity (**FIGURE 2A**). After a

TABLE 1 QUALITY ASSESSMENT OF INCLUDED STUDIES

Study (year)	Adequate definition of case	Representativeness of the cases	Selection of controls	Definition of controls	Comparability of cases and controls	Exposure assessment	Same method of ascertainment for cases and controls	Non-response rate	Total score
<i>Ates et al. (2014)</i>	*			*	*	*	*	*	6
<i>Ağacayak et al. (2016)</i>	*			*	*	*	*	*	6
<i>Bozkaya et al. (2020)</i>	*	*	*	*	*	*	*	*	8
<i>Gulhan et al. (2012)</i>	*			*	*	*	*	*	6
<i>Kalantaridou et al. (2004)</i>	*	*		*	*	*	*	*	7
<i>Kebapcilar et al. (2013)</i>	*	*	*	*	*	*	*	*	8
<i>Knauff et al. (2008)</i>	*	*	*	*	*	*	*	*	8
<i>Kulaksizoglu et al. (2013)</i>	*	*	*	*	**	*	*	*	9
<i>Podfigurna-Stopa et al. (2018)</i>	*			*	**	*	*	*	7
<i>Senoz et al. (1996)</i>	*	*	*	*	**	*	*	*	9

TABLE 2 CHARACTERISTICS OF INCLUDED STUDIES FOR TOTAL CHOLESTEROL

Study (year)	Premature ovarian insufficiency				Control			
	Age, years	BMI (kg/m ²)	n	Total cholesterol, mg/dl	Age, years	BMI, kg/m ²	n	Total cholesterol, mg/dl
	(mean ± SD)	(mean ± SD)		(mean ± SD)	(mean ± SD)	(mean ± SD)		(mean ± SD)
<i>Ates et al. (2014)</i>	35.2 ± 4.6	25.8 ± 4.1	56	191.7 ± 33.3	35.4 ± 4.5	26.0 ± 3.8	59	174.9 ± 37.2
<i>Ağaçayak et al. (2016)</i>	28.9 ± 6.8	24.1 ± 4.2	30	184.5 ± 25.4	29.2 ± 5.0	23.2 ± 3.3	30	170.5 ± 46.9
<i>Bozkaya et al. (2020)</i>	30.9 ± 5.6	25.9 ± 4.2	34	182.9 ± 39.0	28.5 ± 5.1	24.5 ± 3.9	35	167.0 ± 37.5
<i>Gulhan et al. (2012)</i>	36.8 ± 1.8	25.9 ± 4.9	47	200.5 ± 41.8	36.0 ± 2.3	24.8 ± 4.6	60	179.3 ± 32.9
<i>Kalantaridou et al. (2004)</i>	35.4 ± 5.5	24.2 ± 3.1	18	218.0 ± 44.0	35.0 ± 3.6	23.0 ± 2.6	20	202.0 ± 38.0
<i>Kebapcilar et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.5	43	193.3 ± 38.2	37.1 ± 1.8	29.6 ± 3.4	33	172.1 ± 37.1
<i>Knauff et al. (2008)</i>	33.8 ± 5.6	24.9 ± 4.3	90	185.3 ± 36.7	30.3 ± 2.9	24.3 ± 4.8	198	181.5 ± 33.2
<i>Kulaksizoglu et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.6	43	189.3 ± 33.1	37.2 ± 1.9	29.6 ± 3.5	33	168.6 ± 31.6
<i>Podfigurna-Stopa et al. (2018)</i>	30.7 ± 6.9	23.5 ± 4.6	56	203.0 ± 32.4	27.3 ± 4.5	28.6 ± 5.3	68	163.7 ± 17.0
<i>Senoz et al. (1996)</i>	32.8 ± 7.8	26.9 ± 4.8	41	201.0 ± 41.6	27.2 ± 2.0	20.3 ± 2.0	15	177.0 ± 16.7

BMI, body mass index.

case-by-case evaluation, one study was identified and removed (*Podfigurna et al., 2018*). Re-analysis of results showed that no significant heterogeneity was observed ($I^2 = 0\%$, $P = 0.43$). Therefore, a fixed-effects model was applied to evaluate the association between serum total cholesterol levels and POI (nine studies: 402 patients with POI; 483 controls). As shown in **FIGURE 2B**, the results of the meta-analysis showed that serum total cholesterol levels were significantly elevated in patients with POI compared with controls (SMD = 0.39, 95% CI 0.25 to 0.52, $Z = 5.50$, $P < 0.00001$). Among 10 included studies, one study did not examine the levels of serum HDL-C and LDL-C in patients with POI. Therefore, nine included studies were processed for the meta-analysis to evaluate the

changes of serum HDL-C and LDL-C levels between controls and patients with POI. Significant heterogeneity was detected in the HDL-C data ($I^2 > 50\%$, $P < 0.00001$). Unfortunately, after a case-by-case evaluation, sensitivity analyses were unable to identify any study that was the cause of the heterogeneity. Therefore, a random-effects model was applied to examine the association between serum HDL-C levels and POI (nine studies: 428 patients with POI; 521 controls). As shown in **FIGURE 3A**, the meta-analysis results revealed that serum HDL-C levels were not significantly varied between controls and patients with POI (SMD = 0.23, 95% CI -0.13 to 0.59, $Z = 1.26$, $P = 0.21$). Similarly, significant heterogeneity was observed in the LDL-C data ($I^2 > 50\%$, $P = 0.02$). After a case-

by-case evaluation, sensitivity analysis results identified that one study was the cause of the heterogeneity (*Podfigurna et al., 2018*). After removing that study, the results of the meta-analysis with a fixed-effects model (eight studies: 372 POI patients; 453 controls) showed that serum LDL-C levels were significantly increased in patients with POI compared with control patients (SMD = 0.30, 95% CI 0.16 to 0.45, $Z = 4.16$, $P < 0.0001$) (**FIGURE 3B**). Begg's funnel plot test was applied to assess the publication bias. No significant publication bias was detected.

Serum triglyceride levels in controls and patients with premature ovarian insufficiency

Because no significant heterogeneity ($I^2 = 33\%$, $P = 0.15$) was detected in

TABLE 3 CHARACTERISTICS OF INCLUDED STUDIES FOR HIGH-DENSITY LIPOPROTEIN CHOLESTEROL

Study (year)	Premature ovarian insufficiency				Control			
	Age, year	BMI, kg/m ²	n	HDL-C, mg/dl	Age, years	BMI, kg/m ²	n	HDL-C, mg/dl
	(mean ± SD)	(mean ± SD)		(mean ± SD)	(mean ± SD)	(mean ± SD)		(mean ± SD)
<i>Ates et al. (2014)</i>	35.2 ± 4.6	25.8 ± 4.1	56	60.9 ± 16.4	35.4 ± 4.5	26.0 ± 3.8	59	53.9 ± 12.0
<i>Bozkaya et al. (2020)</i>	30.9 ± 5.6	25.9 ± 4.2	34	56.1 ± 10.3	28.5 ± 5.1	24.5 ± 3.9	35	58.4 ± 14.7
<i>Gulhan et al. (2012)</i>	36.8 ± 1.8	25.9 ± 4.9	47	55.5 ± 14.1	36.0 ± 2.3	24.8 ± 4.6	60	51.6 ± 12.1
<i>Kalantaridou et al. (2004)</i>	35.4 ± 5.5	24.2 ± 3.1	18	57.0 ± 17.0	35.0 ± 3.6	23.0 ± 2.6	20	55.0 ± 10.0
<i>Kebapcilar et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.5	43	53.7 ± 11.0	37.1 ± 1.8	29.6 ± 3.4	33	47.5 ± 7.3
<i>Knauff et al. (2008)</i>	33.8 ± 5.6	24.9 ± 4.3	90	57.1 ± 18.5	30.3 ± 2.9	24.3 ± 4.8	198	57.5 ± 12.0
<i>Kulaksizoglu et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.6	43	53.8 ± 11.1	37.2 ± 1.9	29.6 ± 3.5	33	47.5 ± 7.3
<i>Podfigurna-Stopa et al. (2018)</i>	30.7 ± 6.9	23.5 ± 4.6	56	74.0 ± 18.7	27.3 ± 4.5	28.6 ± 5.3	68	57 ± 14.2
<i>Senoz et al. (1996)</i>	32.8 ± 7.8	26.9 ± 4.8	41	51.9 ± 13.0	27.2 ± 2.0	20.3 ± 2.0	15	66.6 ± 7.0

BMI, body mass index; HDL-C, high-density lipoprotein cholesterol.

TABLE 4 CHARACTERISTICS OF INCLUDED STUDIES FOR LOW-DENSITY LIPOPROTEIN CHOLESTEROL

Study (year)	Premature ovarian insufficiency				Control			
	Age, years	BMI, kg/m ²	n	LDL-C, mg/dl	Age, years	BMI, kg/m ²	n	LDL-C, mg/dl
	(mean ± SD)	(mean ± SD)		(mean ± SD)	(mean ± SD)	(mean ± SD)		(mean ± SD)
<i>Ates et al. (2014)</i>	35.2 ± 4.6	25.8 ± 4.1	56	115.4 ± 30.4	35.4 ± 4.5	26.0 ± 3.8	59	109.3 ± 28.5
<i>Bozkaya et al. (2020)</i>	30.9 ± 5.6	25.9 ± 4.2	34	104.6 ± 37.1	28.5 ± 5.1	24.5 ± 3.9	35	88.7 ± 31.3
<i>Gulhan et al. (2012)</i>	36.8 ± 1.8	25.9 ± 4.9	47	118.2 ± 33.2	36.0 ± 2.3	24.8 ± 4.6	60	106.2 ± 26.2
<i>Kalantaridou et al. (2004)</i>	35.4 ± 5.5	24.2 ± 3.1	18	146.0 ± 33.0	35.0 ± 3.6	23.0 ± 2.6	20	128.0 ± 34.0
<i>Kebapcilar et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.5	43	114.3 ± 33.6	37.1 ± 1.8	29.6 ± 3.4	33	99.4 ± 26.6
<i>Knauff et al. (2008)</i>	33.8 ± 5.6	24.9 ± 4.3	90	110.4 ± 33.2	30.3 ± 2.9	24.3 ± 4.8	198	108.5 ± 31.3
<i>Kulaksizoglu et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.6	43	114.3 ± 33.7	37.2 ± 1.9	29.6 ± 3.5	33	99.5 ± 26.6
<i>Podfigurna-Stopa et al. (2018)</i>	30.7 ± 6.9	23.5 ± 4.6	56	114.7 ± 31.2	27.3 ± 4.5	28.6 ± 5.3	68	91.7 ± 15.2
<i>Senoz et al. (1996)</i>	32.8 ± 7.8	26.9 ± 4.8	41	113.0 ± 47.4	27.2 ± 2.0	20.3 ± 2.0	15	80.0 ± 40.1

LDL-C, low-density lipoprotein cholesterol.

serum triglyceride data obtained from all included studies, a fixed-effects model was applied to evaluate the serum triglyceride levels in controls and patients with POI (10 studies: 458 POI patients; 551 controls). Our meta-analysis results showed that serum triglyceride levels in patients with POI were significantly higher than that in control patients (SMD = 0.16, 95% CI 0.03 to 0.29, Z = 2.44, P = 0.01) (FIGURE 4). No significant publication bias was detected by Begg's funnel plot test.

DISCUSSION

Previous studies comparing the serum lipid profiles in patients with and without POI yielded contradictory results. In the present study, we conducted meta-

analyses to examine the changes of serum triglyceride, HDL-C and LDL-C levels between controls and patients with POI. Our results showed that serum total cholesterol, triglyceride and LDL-C levels were significantly elevated in patients with POI compared with those in healthy controls. The serum levels of HDL-C, however, were not significantly varied between patients with POI and controls. To the best of our knowledge, this is the first meta-analysis study to evaluate and summarize the changes in serum lipid profiles between healthy women and women with POI.

Premature ovarian insufficiency is not only a gynaecological problem but is also associated with other cardiovascular diseases. Women with POI have

been reported to have shortened life expectancy that is mainly attributed to the high susceptibility of cardiovascular diseases (*Podfigurna-Stopa et al., 2016*). The risk of mortality from ischaemic heart disease is significantly increased in women with POI compared with women going through menopause at age of 49–55 years (*Jacobsen et al., 1999*). It is well characterized that the abnormal lipid profile is the major risk factor for different cardiovascular diseases and is associated with the development and progression of those diseases. In addition to abnormal serum lipid profile, several risk factors, such as endothelial dysfunction, insulin resistance, cardiovascular autonomic dysfunction and metabolic disorders are observed in patients with POI and

TABLE 5 CHARACTERISTICS OF INCLUDED STUDIES FOR TRIGLYCERIDE

Study (year)	Premature ovarian insufficiency				Control			
	Age, years	BMI, kg/m ²	n	Triglyceride, mg/dl	Age, year	BMI, kg/m ²	n	Triglyceride, mg/dl
	(mean ± SD)	(mean ± SD)		(mean ± SD)	(mean ± SD)	(mean ± SD)		(mean ± SD)
<i>Ates et al. (2014)</i>	35.2 ± 4.6	25.8 ± 4.1	56	84.5 ± 42.4	35.4 ± 4.5	26.0 ± 3.8	59	94.1 ± 95.3
<i>Ağaayak et al. (2016)</i>	28.9 ± 6.8	24.1 ± 4.2	30	124.4 ± 25.6	29.2 ± 5.0	23.2 ± 3.3	30	110.8 ± 37.0
<i>Bozkaya et al. (2020)</i>	30.9 ± 5.6	25.9 ± 4.2	34	130.1 ± 123.4	28.5 ± 5.1	24.5 ± 3.9	35	97.8 ± 36.5
<i>Gulhan et al. (2012)</i>	36.8 ± 1.8	25.9 ± 4.9	47	128.5 ± 81.2	36.0 ± 2.3	24.8 ± 4.6	60	102.4 ± 54.1
<i>Kalantaridou et al. (2004)</i>	35.4 ± 5.5	24.2 ± 3.1	18	73.0 ± 27.0	35.0 ± 3.6	23.0 ± 2.6	20	95.0 ± 36.0
<i>Kebapcilar et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.5	43	157.9 ± 97.1	37.1 ± 1.8	29.6 ± 3.4	33	142.5 ± 76.4
<i>Knauff et al. (2008)</i>	33.8 ± 5.6	24.9 ± 4.3	90	91.2 ± 38.1	30.3 ± 2.9	24.3 ± 4.8	198	81.4 ± 43.4
<i>Kulaksizoglu et al. (2013)</i>	36.8 ± 2.7	30.3 ± 5.6	43	137.9 ± 59.8	37.2 ± 1.9	29.6 ± 3.5	33	119.1 ± 40.6
<i>Podfigurna-Stopa et al. (2018)</i>	30.7 ± 6.9	23.5 ± 4.6	56	77.3 ± 48.5	27.3 ± 4.5	28.6 ± 5.3	68	76.1 ± 26.9
<i>Senoz et al. (1996)</i>	32.8 ± 7.8	26.9 ± 4.8	41	141.7 ± 80.7	27.2 ± 2.0	20.3 ± 2.0	15	143.2 ± 133.4

BMI, body mass index.

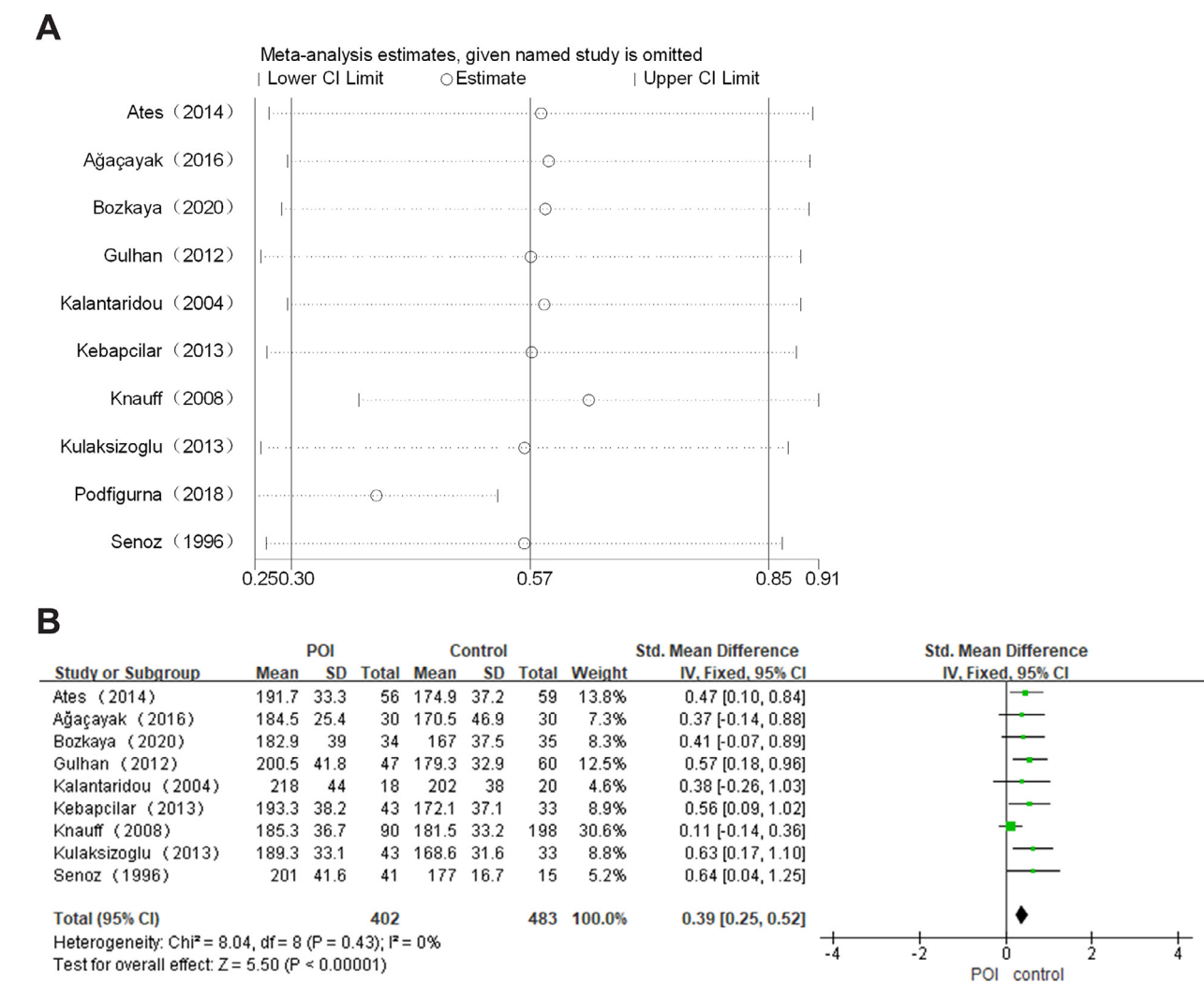


FIGURE 2 (A) Sensitivity analysis of serum total cholesterol levels; (B) overall meta-analysis result of serum total cholesterol levels in patients with premature ovarian insufficiency (POI) and controls.

have been reported to contribute to the pathogenesis of cardiovascular diseases (Podfigurna-Stopa et al., 2016).

Most cardiovascular diseases are attributable to the raised serum total cholesterol levels. A previous meta-analysis study shows that serum total cholesterol is a risk factor for cardiovascular diseases in women and men (Peters et al., 2016). Although serum total cholesterol levels play an important role in the risk assessment of cardiovascular diseases, individual components such as HDL-C and LDL-C have been shown to play a more integral role in the development and progression of cardiovascular diseases. HDL-C is considered to be good cholesterol because high HDL-C levels are strongly associated with a low risk of cardiovascular diseases.

In contrast, LDL-C is known as bad cholesterol because it carries cholesterol to tissues and plays a major role in the pathophysiology of atherosclerotic cardiovascular diseases. Compared with age-matched men, women of reproductive age have higher serum HDL-C and lower LDL-C levels. After the menopause, however, serum HDL-C levels decline and LDL-C levels rise (Campos et al., 1988; Stevenson et al., 1993). It has been reported that oral administration of oestrogen reduces serum total cholesterol and LDL-C levels and increases HDL-C levels in postmenopausal women (Fahraeus, 1988; Lobo, 1991; Walsh et al., 1991). Because of the protective effects of oestrogen during a woman's reproductive years, cardiovascular diseases occur at a much lower rate in women of reproductive age compared with men of the same age

(Guetta and Cannon, 1996). Attrition in the number of ovarian follicles and ensuing insufficiency in ovarian sex hormone productions, particularly for the oestrogen, are hallmarks of POI (Welt, 2008). Therefore, the elevated serum levels of total cholesterol and LDL-C in patients with POI reported by the present study could be attributed to the deficiency of oestrogen production. Our meta-analysis results showed that serum HDL-C levels were not significantly varied between controls and patients with POI. In postmenopausal women, however, the decline of endogenous oestrogen does not seem to affect serum HDL-C levels (Fahraeus, 1988). In addition, the effect of oestrogen supplements on serum HDL-C levels remains controversial. Collectively, these findings suggest that serum HDL-C levels are not mainly regulated by endogenous oestrogen, which could

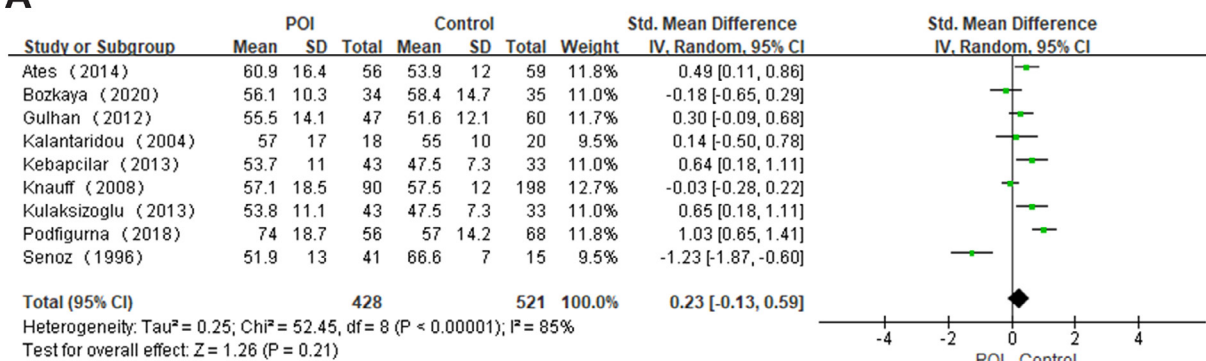
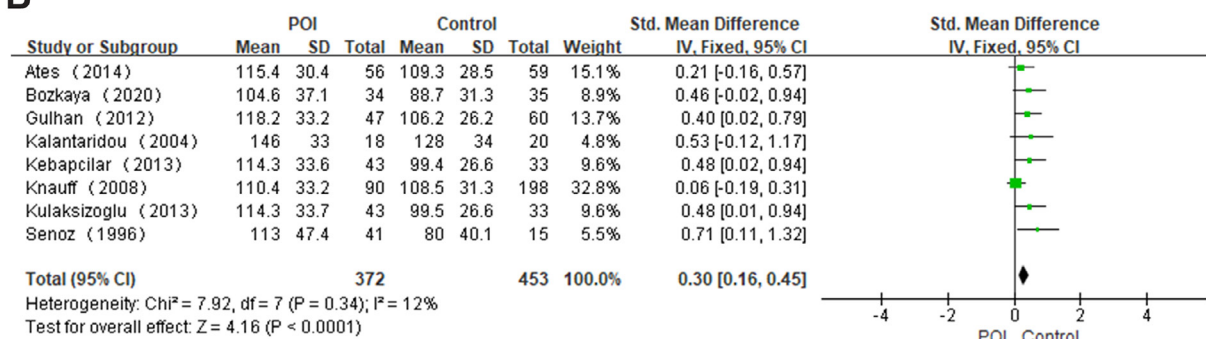
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FIGURE 3 (A) Overall meta-analysis result of serum high-density lipoprotein cholesterol levels in patients with premature ovarian insufficiency (POI) and controls; (B) overall meta-analysis result of serum low-density lipoprotein cholesterol levels in patients with POI and controls.

explain the results that serum levels of HDL-C were similar in patients with POI and controls.

Studies on the effects of oestrogen administration on serum lipid profile in postmenopausal women have produced contradictory results depending on the route of administration (*Guetta and Cannon, 1996*). Orally administered oestrogen decreases serum LDL-C levels and elevates serum HDL-C levels. Transdermal oestrogen administration, however, has statistically and clinically non-significant effects on serum

lipoprotein levels but can improve vascular endothelial function (*Walsh et al., 1991; Moreau et al., 2013*). Regardless of the negative results of hormone replacement therapy (HRT) for postmenopausal women (*Cagnacci and Venier, 2019*), HRT is strongly recommended by the European Society of Human Reproduction and Embryology for women with POI, although studies investigating optimal HRT for POI remain limited (*ESHRE et al., 2016*). Hormone replacement therapy can relieve vasomotor and genitourinary symptoms in patients with POI. In addition, HRT

helps to maintain bone health, restore endothelial dysfunction and decrease the risk and mortality of cardiovascular diseases (*Webber et al., 2017*). To date, however, the effect of HRT on serum lipid profile in patients with POI remains undetermined and warrants further investigations.

Although conflicting data have been reported, insulin resistance commonly occurs in patients with POI (*Ishizuka, 2021*). Triglyceride is directly lipolysed by lipoprotein lipase. It is known that insulin resistance decreases lipoprotein lipase

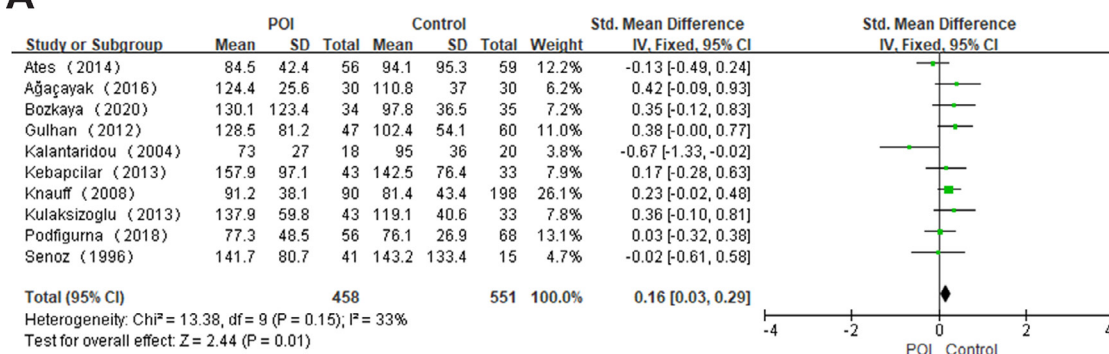
A

FIGURE 4 Overall meta-analysis result of serum triglyceride levels in patients with premature ovarian insufficiency (POI) and controls.

activity, which results in elevated serum triglyceride levels. Many studies have demonstrated that serum triglyceride levels are an independent risk factor for cardiovascular diseases (*Harchaoui, et al., 2009*). During menopause, serum triglyceride levels are also increased owing to the increase of the abdominal fat amount and decrease in insulin sensitivity (*Derby et al., 2009*). Our meta-analysis results revealed that serum triglyceride levels were significantly higher in patients with POI than in controls. Interestingly, orally administered oestrogen increases serum triglyceride levels in postmenopausal women (*Godsland, 2001*). These results suggest that elevated serum triglyceride levels in patients with POI is not attributed to the insufficiency of oestrogen. In addition, why serum triglyceride levels are increased in patients with POI is unclear and needs further investigation.

We are aware that not all confounding factors were adjusted in our study. For example, smoking history is not presented in all included studies but has been shown to affect the serum lipid profiles. In addition, although eight studies showed no significant difference in age and BMI between control and patients with POI, two studies showed that age and BMI were statistically different between controls and patients with POI (*Senoz, et al., 1996; Podfigurna et al., 2018*). Moreover, we are aware that the results of our meta-analysis were derived from a limited sample size of included published studies. Therefore, future studies with a large sample size and more included studies are required to evaluate the association between lipid profiles and the risk of POI.

In conclusion, the present meta-analysis shows that serum total cholesterol, LDL-C and triglyceride levels are significantly increased in patients with POI compared with healthy controls. The elevations of these lipids may, in part, be attributed to the ovarian insufficiency, resulting from deficiency of oestrogen. In addition, the abnormal lipid profile contributes to the risk of cardiovascular diseases that frequently occur and lead to high mortality in patients with POI.

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REFERENCES

- Agacayak, E., Yaman Goruk, N., Kusen, H., Yaman Tunc, S., Basaranoglu, S., Icen, M.S., Yildizbakan, A., Yuksel, H., Kalkanli, S., Gul, T. **Role of inflammation and oxidative stress in the etiology of primary ovarian insufficiency.** Turkish journal of obstetrics and gynecology 2016; 13: 109–115
- Ates, S., Yesil, G., Sevket, O., Molla, T., Yildiz, S. **Comparison of metabolic profile and abdominal fat distribution between karyotypically normal women with premature ovarian insufficiency and age matched controls.** Maturitas 2014; 79: 306–310
- Begg, C.B., Mazumdar, M. **Operating characteristics of a rank correlation test for publication bias.** Biometrics 1994; 50: 1088–1101
- Bozkaya, V.O., Yumusak, O.H., Ozaksit, G., Tenekecioglu, E., Gul Ibrism, E., Alkan, M., Oskovi-Kaplan, Z.A., Erel, O. **The role of oxidative stress on subclinical atherosclerosis in premature ovarian insufficiency and relationship with carotid intima-media thickness.** Gynecological endocrinology: the official journal of the International Society of Gynecological Endocrinology 2020; 36: 687–692
- Cagnacci, A., Venier, M. **The Controversial History of Hormone Replacement Therapy.** Medicina 2019; 55
- Campos, H., McNamara, J.R., Wilson, P.W., Ordovas, J.M., Schaefer, E.J. **Differences in low density lipoprotein subfractions and apolipoproteins in premenopausal and postmenopausal women.** The Journal of clinical endocrinology and metabolism 1988; 67: 30–35
- Cohen, J. 1988 **Statistical power analysis for the behavioral sciences.** 2nd edn L. Erlbaum Associates Hillsdale, N.J.
- Deeks, J.J., Dinnes, J., D'Amico, R., Sowden, A.J., Sakarovich, C., Song, F., Petticrew, M., Altman, D.G. **International Stroke Trial Collaborative G., European Carotid Surgery Trial Collaborative G. Evaluating non-randomised intervention studies.** Health technology assessment 2003; 7: 1–173
- Derby, C.A., Crawford, S.L., Pasternak, R.C., Sowers, M., Sternfeld, B., Matthews, K.A. **Lipid changes during the menopause transition in relation to age and weight: the Study of Women's Health Across the Nation.** American journal of epidemiology 2009; 169: 1352–1361
- ESHRE. **Embryology Guideline Group on P.O.I.** Webber, L., Davies, M., Anderson, R., Bartlett, J., Braat, D., Cartwright, B., Cifkova, R., de Muinck Keizer-Schrama, S., Hogervorst, E., Janse, F., Liao, L., Vlasisavljevic, V., Zillikens, C., Vermeulen, N. **ESHRE Guideline: management of women with premature ovarian insufficiency.** Human reproduction 2016; 31: 926–937
- Fahraeus, L. **The effects of estradiol on blood lipids and lipoproteins in postmenopausal women.** Obstetrics and gynecology 1988; 72: 18S–22S
- Godsland, I.F. **Effects of postmenopausal hormone replacement therapy on lipid, lipoprotein, and apolipoprotein (a) concentrations: analysis of studies published from 1974–2000.** Fertility and sterility 2001; 75: 898–915

- Goswami, D., Conway, G.S. **Premature ovarian failure.** Human reproduction update 2005; 11: 391–410
- Guetta, V., Cannon, R.O. **Cardiovascular effects of estrogen and lipid-lowering therapies in postmenopausal women.** Circulation 1996; 93: 1928–1937
- Gulhan, I., Bozkaya, G., Uyar, I., Oztekin, D., Pamuk, B.O., Dogan, E. **Serum lipid levels in women with premature ovarian failure.** Menopause 2012; 19: 1231–1234
- Gunning, M.N., Meun, C., van Rijn, B.B., Daan, N.M.P., Roeters van Lennep, J.E., Appelman, Y., Boersma, E., Hofstra, L., Fauser, C., Rueda-Ochoa, O.L., Ikram, M.A., Kavousi, M., Lambalk, C.B., Eijkemans, M.J.C., Laven, J.S.E., Fauser, B. **consortium C. The cardiovascular risk profile of middle age women previously diagnosed with premature ovarian insufficiency: A case-control study.** PLoS one 2020; 15e0229576
- Gunning, M.N., Meun, C., van Rijn, B.B., Maas, A., Benschop, L., Franx, A., Boersma, E., Budde, R.P.J., Appelman, Y., Lambalk, C.B., Eijkemans, M.J.C., Velthuis, B.K., Laven, J.S.E. **Fausser B., consortium C. Coronary artery calcification in middle-aged women with premature ovarian insufficiency.** Clinical endocrinology 2019; 91: 314–322
- Harchaoui, K.E., Visser, M.E., Kastelein, J.J., Stroes, E.S., Dallinga-Thie, G.M. **Triglycerides and cardiovascular risk.** Current cardiology reviews 2009; 5: 216–222
- Higgins, J., Thompson, S., Deeks, J., Altman, D. **Statistical heterogeneity in systematic reviews of clinical trials: a critical appraisal of guidelines and practice.** Journal of health services research & policy 2002; 7: 51–61
- Higgins, J.P.T., Green, S. **Cochrane Collaboration. 2008 Cochrane handbook for systematic reviews of interventions.** Wiley-Blackwell/Chichester, England; Hoboken, NJ
- Ishizuka, B. **Current Understanding of the Etiology, Symptomatology, and Treatment Options in Premature Ovarian Insufficiency (POI).** Frontiers in endocrinology 2021; 12e26924
- Jacobsen, B.K., Knutsen, S.F., Fraser, G.E. **Age at natural menopause and total mortality and mortality from ischemic heart disease: the Adventist Health Study.** Journal of clinical epidemiology 1999; 52: 303–307
- Johansen, M., Thomsen, S.F. **Guidelines for Reporting Medical Research: A Critical Appraisal.** International scholarly research notices 2016; 20161346026
- Kalantaridou, S.N., Naka, K.K., Papanikolaou, E., Kazakos, N., Kravariti, M., Calis, K.A., Paraskevaidis, E.A., Sideris, D.A., Tsatsoulis, A., Chrousos, G.P., Michalis, L.K. **Impaired endothelial function in young women with premature ovarian failure: normalization with hormone therapy.** The Journal of clinical endocrinology and metabolism 2004; 89: 3907–3913
- Kebapcilar, A.G., Kulaksizoglu, M., Ipekci, S.H., Korkmaz, H., Kebapcilar, L., Akyurek, F., Taner, C.E., Gonen, M.S. **Relationship between mean platelet volume and low-grade systemic coagulation with vitamin D deficiency in primary ovarian insufficiency.** Archives of gynecology and obstetrics 2013; 288: 207–212
- Knauff, E.A., Westerveld, H.E., Goverde, A.J., Eijkemans, M.J., Valkenburg, O., van Santbrink, E.J., Fauser, B.C., van der Schouw, Y.T. **Lipid profile of women with premature ovarian failure.** Menopause 2008; 15: 919–923
- Kulaksizoglu, M., Ipekci, S.H., Kebapcilar, L., Kebapcilar, A.G., Korkmaz, H., Akyurek, F., Baldane, S., Gonen, M.S. **Risk factors for diabetes mellitus in women with primary ovarian insufficiency.** Biological trace element research 2013; 154: 313–320
- Lemieux, I., Lamarche, B., Couillard, C., Pascot, A., Cantin, B., Bergeron, J., Dagenais, G.R., Despres, J.P. **Total cholesterol/HDL cholesterol ratio vs LDL cholesterol/HDL cholesterol ratio as indices of ischemic heart disease risk in men: the Quebec Cardiovascular Study.** Archives of internal medicine 2001; 161: 2685–2692
- Lobo, R.A. **Clinical review 27: Effects of hormonal replacement on lipids and lipoproteins in postmenopausal women.** The Journal of clinical endocrinology and metabolism 1991; 73: 925–930
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., Group, P. **Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.** PLoS medicine 2009; 6e1000097
- Moreau, K.L., Stauffer, B.L., Kohrt, W.M., Seals, D.R. **Essential role of estrogen for improvements in vascular endothelial function with endurance exercise in postmenopausal women.** The Journal of clinical endocrinology and metabolism 2013; 98: 4507–4515
- Peters, S.A., Singhathe, Y., Mackay, D., Huxley, R.R., Woodward, M. **Total cholesterol as a risk factor for coronary heart disease and stroke in women compared with men: A systematic review and meta-analysis.** Atherosclerosis 2016; 248: 123–131
- Pirruccello, J., Kathiresan, S. **Genetics of lipid disorders.** Current opinion in cardiology 2010; 25: 238–242
- Podfigurna-Stopa, A., Czyzyk, A., Grymowicz, M., Smolarczyk, R., Katulski, K., Czajkowski, K., Meczekalski, B. **Premature ovarian insufficiency: the context of long-term effects.** Journal of endocrinological investigation 2016; 39: 983–990
- Podfigurna, A., Stellmach, A., Szeliga, A., Czyzyk, A., Meczekalski, B. **Metabolic Profile of Patients with Premature Ovarian Insufficiency.** Journal of clinical medicine 2018; 7
- Rivera, C.M., Grossardt, B.R., Rhodes, D.J., Brown, R.D.Jr., Roger, V.L., Melton, L.J. **3rd, Rocca, W.A. Increased cardiovascular mortality after early bilateral oophorectomy.** Menopause 2009; 16: 15–23
- Rocca, W.A., Grossardt, B.R., de Andrade, M., Malkasian, G.D., Melton, L.J. **3rd Survival patterns after oophorectomy in premenopausal women: a population-based cohort study.** The Lancet Oncology 2006; 7: 821–828
- Roeters van Lennep, J.E., Heida, K.Y., Bots, M.L., Hoek, A. **collaborators of the Dutch Multidisciplinary Guideline Development Group on Cardiovascular Risk Management after Reproductive D. Cardiovascular disease risk in women with premature ovarian insufficiency: A systematic review and meta-analysis.** European journal of preventive cardiology 2016; 23: 178–186
- Senoz, S., Direm, B., Gulekli, B., Gokmen, O. **Estrogen deprivation, rather than age, is responsible for the poor lipid profile and carbohydrate metabolism in women.** Maturitas 1996; 25: 107–114
- Stang, A. **Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses.** European journal of epidemiology 2010; 25: 603–605
- Stevenson, J.C., Crook, D., Godsland, I.F. **Influence of age and menopause on serum lipids and lipoproteins in healthy women.** Atherosclerosis 1993; 98: 83–90
- Takeshima, N., Sozu, T., Tajika, A., Ogawa, Y., Hayasaka, Y., Furukawa, T.A. **Which is more generalizable, powerful and interpretable in meta-analyses, mean difference or standardized mean difference? BMC medical research methodology 2014; 14: 30**
- Taylor, A.E. **Systemic adversities of ovarian failure.** Journal of the Society for Gynecologic Investigation 2001; 8: S7–S9
- Walsh, B.W., Schiff, I., Rosner, B., Greenberg, L., Ravnika, V., Sacks, F.M. **Effects of postmenopausal estrogen replacement on the concentrations and metabolism of plasma lipoproteins.** The New England journal of medicine 1991; 325: 1196–1204
- Webber, L., Anderson, R.A., Davies, M., Janse, F., Vermeulen, N. **HRT for women with premature ovarian insufficiency: a comprehensive review.** Human reproduction open 2017; 2017: hox007
- Welt, C.K. **Primary ovarian insufficiency: a more accurate term for premature ovarian failure.** Clinical endocrinology 2008; 68: 499–509

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